Module 5 — Quantifying Uncertainty

Why do we need probability in AI?

- In classical logic, we often have statements that are either **true or false**.
- For example: "It is raining.“ We might represent: $Rain$
- But in the real world, an AI system often does **not know with certainty** whether something is true.
- **For example:** "Will it rain tomorrow?"
- We cannot simply say: True ❌ or False ❌
- Instead, we might say: There is a **70% chance** of rain.
- So probability allows an AI system to reason when there is **uncertainty**.
- **Example:** A medical AI might observe: fever , cough , fatigue
- It may not be certain that the patient has flu. Instead: $P(Flu) = 0.6$
- meaning: Based on the available information, there is a 60% probability of flu.
- **Probability allows AI to reason about uncertain situations instead of using only True/False logic.**

What is probability?

- **Probability measures how likely an event is to occur.**
- Probability is always between **0 and 1**.
- $0 \leq P(A) \leq 1$, where $A$is an event.
- **Probability of an event :** When all outcomes are equally likely:

$$P(A) = \frac{Number\ of\ favorable\ outcomes}{Total\ number\ of\ outcomes}$$

- **Example 1 —** A fair die is rolled. What is the probability of getting **4**?

  P(4)=1/6

- What is the probability of getting an even number? P(even) = 3/6

- A **probability distribution** is a mathematical function that describes the likelihood of obtaining the possible outcomes of a random variable. It map every possible value of an event to its corresponding probability of occurring, with all total probabilities summing (or integrating) to 1.

Outcome	Probability
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Suppose we roll a fair die. We can assign a probability to **every possible outcome**. The probabilities must add up to 1.

$$\frac{1}{6}+\frac{1}{6}+\frac{1}{6}+\frac{1}{6}+\frac{1}{6}+\frac{1}{6}=1$$

This is an important rule:

$$\boxed{\sum P\,(x)=1}$$

The most important formula is: $P(\neg A)=1-P(A)$

- **Mutually exclusive events :** Two events are **mutually exclusive** if they cannot happen at the same time.
- For example, when rolling a die: getting 2 and getting 5 cannot happen simultaneously in one roll.
- Therefore they are mutually exclusive.
- We write: $A \cap B = \emptyset$
- General addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- For two mutually exclusive events: $P(A \cup B) = P(A) + P(B)$

Important probability rules

- **Rule 1 — Probability range**

$$\boxed{0 \leq P(A) \leq 1}$$

- **Rule 2 — Certain event**

$$\boxed{P(S) = 1}$$

- **Rule 3 — Impossible event**

$$\boxed{P(\emptyset) = 0}$$

- **Rule 4 — Complement**

$$\boxed{P(\neg A) = 1 - P(A)}$$

- **Rule 5 — General addition rule**

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

Example

- A class has 100 students. 60 students know Python. 40 students know Java. 20 students know both Python and Java.
- Choose one student randomly. What is the probability that the student knows **Python or Java**?
- Let: $P =$ knows Python $J =$ knows Java
- Given:
- $P(P) = \frac{60}{100} = 0.6$
- $P(J) = \frac{40}{100} = 0.4$
- $P(P \cap J) = \frac{20}{100} = 0.2$
- Using the addition rule: $P(P \cup J) = P(P) + P(J) - P(P \cap J) = 0.6 + 0.4 - 0.2 = \boxed{0.8}$
- Therefore: $\boxed{80\%}$ of the students know Python or Java.

Conditional Probability

- Conditional probability tells us the probability of an event when we already know that another event has occurred.
- Conditional probability is written as: $\boxed{P(A \mid B)}$
- Read it as:**Probability of A given B**
- For example: $P(Rain \mid Cloudy)$ means: Probability of rain **given that it is cloudy**.

Conditional Probability

- Consider a medical AI.
- Suppose: $P(Flu) = 0.10$
- So, before seeing any symptoms, the probability that a randomly selected person has flu is **10%**.
- Now suppose we learn that the person has a **fever**.
- Does this new information change our belief? Yes.
- Perhaps: $P(Flu \mid Fever) = 0.60$
- This means: **The probability of flu, given that the person has fever, is 60%.**
- The vertical bar | means **"given that"**.

- **Basic formula:** The most important formula in this topic is:

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}$$

- provided: $P(B) > 0$

Remember:

- $P(A \mid B)$→ A **given** B
- $P(A \cap B)$→ A **AND** B
- $P(B)$→ probability of the condition/event we already know

Example

- Suppose there are **100 students**. 40 students are female. 20 female students know Python.
- Question: What is the probability that a student knows Python **given that the student is female**?
- Once we know the student is female, we don't consider all 100 students anymore. We consider only the **40 female students**.
- Among those 40: 20 know Python.
- Therefore: $P(Python \mid Female) = \frac{20}{40} = \boxed{0.5}$ or: $\boxed{50\%}$
- **This is the central intuition: Conditional probability changes the reference group.**
- Instead of asking:
- "Out of all students, how many know Python?"
- we ask:
- "Out of female students, how many know Python?"

Example

- A die is rolled. Let: A= getting an even number and B= getting a number greater than 3
- Find: $P(A \mid B)$
- **Step 1: Sample space** $S = \{1, 2, 3, 4, 5, 6\}$
- **Step 2: Define B** Numbers greater than 3:
- $B = \{4, 5, 6\}$ Therefore: $P(B) = \frac{3}{6} = \frac{1}{2}$
- **Step 3: Find** $A \cap B$ Even numbers: $A = \{2, 4, 6\}$
- Both even **AND** greater than 3: $A \cap B = \{4, 6\}$
- Therefore: $P(A \cap B) = \frac{2}{6} = \frac{1}{3}$
- **Step 4: Apply the formula** $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{1/3}{1/2} = \boxed{\frac{2}{3}}$
- So: $\boxed{P(Even \mid > 3) = \frac{2}{3}}$

Example:

Consider 100 students:

Question: What is the probability that a student knows Python **given that the student is female**?

We already know the student is female.

Therefore our universe becomes:

40 females

Among them:

20 know Python.

Hence:

$$P(Python \mid Female) = \frac{20}{40} = \boxed{0.5}$$

	Python	**Not Python**	**Total**
Male	30	30	60
Female	20	20	40
Total	50	50	**100**

Important points:

- P(A|B) is NOT generally equal to P(B|A)
- **Formula 1 — Conditional probability**

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}$$

- **Formula 2 — Multiplication rule**

$$\boxed{P(A \cap B) = P(A \mid B)P(B)}$$

- **Formula 3 — Reverse form**

$$\boxed{P(A \cap B) = P(B \mid A)P(A)}$$

Joint Probability

- Joint probability = probability that two or more events occur together.
- Joint probability is the probability of **A AND B** occurring together.
- It is written as: $\boxed{P(A \cap B)}$ or sometimes: $\boxed{P(A, B)}$ Read it as:"Probability of A and B."
- **Example**
- Let:
- A= student knows Python
- B= student knows Java
- Then: $P(A \cap B)$ means: Probability that a student knows **both Python AND Java**.

Joint Probability Table

Joint probabilities are often represented using a **table**.

Suppose students are classified based on:

- Gender
- Programming skill

The cell:

$$0.30$$

in the Male/Python cell represents:

$$\boxed{P(Male \cap Python)}$$

Similarly:

$$\boxed{P(Female \cap Python) = 0.20}$$

The entire table is called a **joint probability distribution** for these two variables.

	Python	**Not Python**	**Total**
Male	0.30	0.30	0.60
Female	0.20	0.20	0.40
Total	**0.50**	**0.50**	**1.00**

Marginal Probability

Suppose we have the previous table:
The probability:

$$P(Male) = 0.60$$

is obtained by adding the row:

$$P(Male) = P(Male \cap Python) + P(Male \cap NotPython)$$
$$= 0.30 + 0.30$$
$$= 0.60$$

This is called a **marginal probability**. (This is due to ,when probabilities are written in a table, the totals appear along the **margins**.)
Similarly:

$$P(Python) = 0.30 + 0.20 = 0.50$$

	Python	**Not Python**	**Total**
Male	0.30	0.30	0.60
Female	0.20	0.20	0.40
Total	0.50	0.50	1.00

Joint Probability Distribution

- Suppose we have two random variables: $Gender$ and $Programming$
- A **joint probability distribution** tells us the probability for **every combination**.
- For example:

$$P(Male, Python) = 0.30$$
$$P(Male, NotPython) = 0.30$$
$$P(Female, Python) = 0.20$$
$$P(Female, NotPython) = 0.20$$

- All these probabilities together form the joint distribution.
- And importantly: $0.30 + 0.30 + 0.20 + 0.20 = 1$

Joint Probability and Conditional Probability

- There is a very useful relationship:

$$\boxed{P(A \cap B) = P(A \mid B)P(B)}$$

- Suppose: $P(B) = 0.4$ and: $P(A \mid B) = 0.7$
- Then: $P(A \cap B) = 0.7 \times 0.4 = \boxed{0.28}$
- So the joint probability is **28%**.
- Joint probability and conditional probability are closely connected.

Example: Suppose 200 students are classified according to whether they know Python and Java. Find:

(a) $P(Python \cap Java)$**:** There are 80 students who know both. $P(Python \cap Java) = \frac{80}{200} = \boxed{0.40}$

(b) $P(Python)$

Total Python students: $80 + 40 = 120$

Therefore: $P(Python) = \frac{120}{200} = \boxed{0.60}$

(c) $P(Java)$

Total Java students: $80 + 30 = 110$

Therefore: $P(Java) = \frac{110}{200} = \boxed{0.55}$

(d) $P(Java \mid Python)$**:** Among Python students, there are: 120

Among them, 80 know Java.

Therefore: $P(Java \mid Python) = \frac{80}{120} = = \boxed{0.667}$

	Java	**Not Java**	**Total**
Python	80	40	120
Not Python	30	50	80
Total	110	90	200

Bayes Rule

- Bayes Rule allows us to update our belief about a hypothesis when we receive new evidence.

- Consider a medical AI.
- Suppose: A disease is relatively rare. A patient gets a positive medical test.
- The important question is: **What is the probability that the patient actually has the disease, given that the test is positive?**
- Mathematically: $P(Disease \mid Positive)$
- But often, what we know from medical studies is: $P(Positive \mid Disease)$
- These are **not the same thing**.
- Bayes Rule gives us a way to calculate: $\boxed{P(Disease \mid Positive)}$ using information such as: $P(Positive \mid Disease)$

Deriving Bayes Rule

- We already learned the multiplication rule.

$$P(A \cap B) = P(A \mid B)P(B)$$

- We can also write:

$$P(A \cap B) = P(B \mid A)P(A)$$

- Since both expressions represent the same joint probability:

$$P(A \mid B)P(B) = P(B \mid A)P(A)$$

- Divide both sides by $P(B)$:

$$\boxed{P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}}$$

- This is **Bayes Rule**.

The four components of Bayes Rule

$$\boxed{P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}}$$

or

$$\boxed{Posterior = \frac{Likelihood \times Prior}{Evidence}}$$

Term	Name	Meaning
$P(A)$	Prior	Initial belief about A
$P(B \mid A)$	Likelihood	Probability of evidence B if A is true
$P(B)$	Evidence / Marginal likelihood	Overall probability of B
$P(A \mid B)$	Posterior	Updated belief about A after seeing B

Example 1: Medical Testing

- Imagine a rare disease that affects **1% of the population**. You take a medical test for it:
- The test is **90% accurate** if you have the disease (True Positive Rate: $P(\text{Positive}|\text{Sick}) = 0.90$).
- The test has a **10% false positive rate** if you don't have it ($P(\text{Positive}|\text{Healthy}) = 0.10$).
- If you test **positive**, what is the actual probability that you are sick?

State the Formula

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Map the Variables to the Problem

- Let A be the event that you are **Sick**.
- Let A^c be the event that you are **Healthy** (Not Sick).
- Let B be the event that you test **Positive**.
- We want to find $\boldsymbol{P}(\textbf{Sick}|\textbf{Positive})$.

Given values:

- **Prior probability of being sick,** $P(A)$**:** 0.01
- **Prior probability of being healthy,** $P(A^c)$**:** $1 - 0.01 = 0.99$
- **Likelihood of testing positive if sick,** $P(B|A)$**:** 0.90
- **Likelihood of testing positive if healthy,** $P(B|A^c)$**:** 0.10

Calculate $P(B)$**(Total Probability of Testing Positive)**

- Since a positive test can come from either a sick person or a healthy person, we use the Law of Total Probability to find $P(B)$:

$$P(B) = [P(B|A) \cdot P(A)] + [P(B|A^c) \cdot P(A^c)]$$

- Substitute the given numbers:

$$P(B) = (0.90 \cdot 0.01) + (0.10 \cdot 0.99)$$
$$P(B) = 0.009 + 0.099 = 0.108$$

Apply Bayes' Rule

- Now, plug $P(A)$, $P(B|A)$, and $P(B)$directly into the main formula:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$P(\text{Sick}|\text{Positive}) = \frac{0.90 \cdot 0.01}{0.108}$$

$$P(\text{Sick}|\text{Positive}) = \frac{0.009}{0.108}$$

$$P(\text{Sick}|\text{Positive}) = \frac{1}{12} \approx 0.0833 \text{ or } 8.33\%$$

How P(B) Computed?

- **Step 1: Break Down Event B(Testing Positive)**
- Anyone who receives a positive test (B) must fall into one of two mutually exclusive groups: They are **Sick AND test Positive** ($A \cap B$). They are **Healthy AND test Positive** ($A^c \cap B$).
- Because these two groups do not overlap, the total probability of testing positive, $P(B)$, is simply the sum of both probabilities: $P(B) = P(A \cap B) + P(A^c \cap B)$
- **Step 2: Express Joint Probabilities using Conditional Probability**
- The definition of conditional probability states: $P(B|A) = \frac{P(A \cap B)}{P(A)}$
- If you multiply both sides by $P(A)$, you get the formula for the probability of both events occurring together: $P(A \cap B) = P(B|A) \cdot P(A)$
- This gives us:
- **Probability of being Sick AND testing Positive:** $P(A \cap B) = P(B|A) \cdot P(A)$
- **Probability of being Healthy AND testing Positive:** $P(A^c \cap B) = P(B|A^c) \cdot P(A^c)$
- **Step 3: Substitute Back into $P(B)$**
- Substituting these two parts back into the equation from Step 1 gives the final formula:

$$P(B) = [P(B|A) \cdot P(A)] + [P(B|A^c) \cdot P(A^c)]$$

Example 2: Spam Email Filter

- Suppose your email filter is trying to determine if an email containing the word **"FREE"** is spam.
- **Prior belief:** 20% of all emails you receive are spam ($P(\text{Spam}) = 0.20$).
- **Likelihood:** 80% of spam emails contain the word "FREE" ($P(\text{"FREE"}|\text{Spam}) = 0.80$).
- **Baseline word frequency:** 10% of legitimate (non-spam) emails also contain the word "FREE" ($P(\text{"FREE"}|\text{Legit}) = 0.10$).
- What is the probability an email is spam if it contains the word "FREE"?

- **Calculate total probability of seeing "FREE" ($P(B)$):**

$$P(\text{"FREE"}) = [P(\text{"FREE"}|\text{Spam}) \cdot P(\text{Spam})] + [P(\text{"FREE"}|\text{Legit}) \cdot P(\text{Legit})]$$

$$P(\text{"FREE"}) = (0.80 \cdot 0.20) + (0.10 \cdot 0.80) = 0.16 + 0.08 = 0.24$$

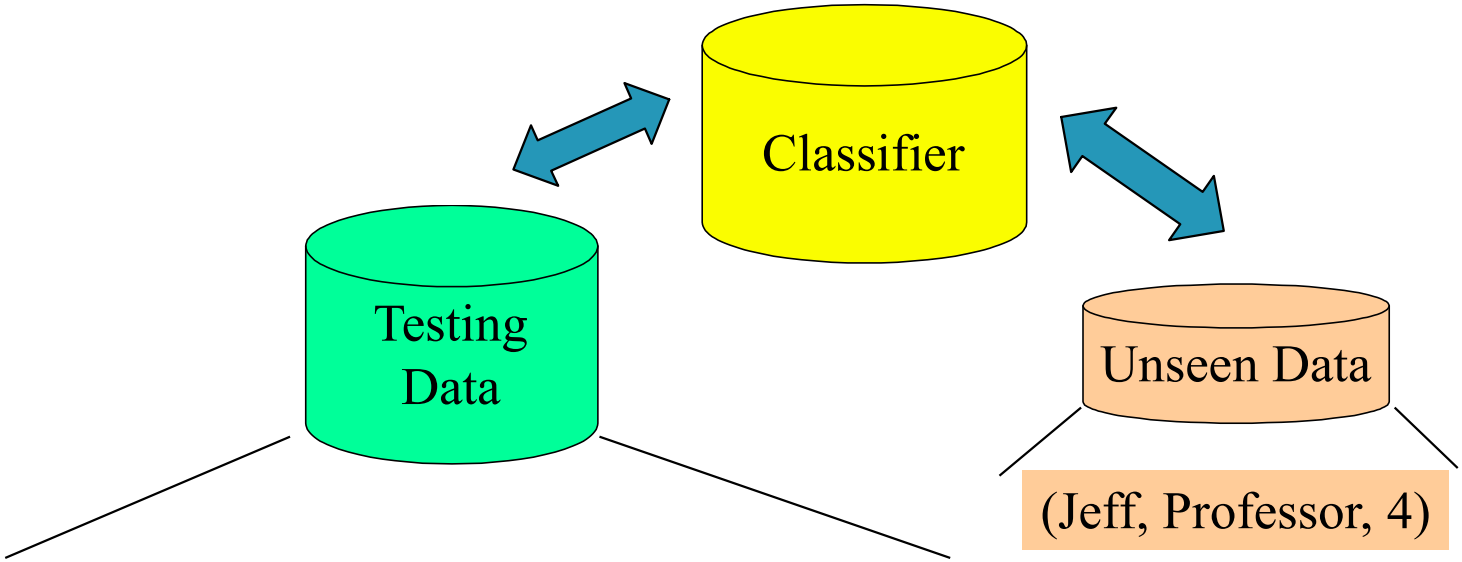
- **Apply Bayes' Rule:**

$$P(\text{Spam}|\text{"FREE"}) = \frac{0.80 \cdot 0.20}{0.24} = \frac{0.16}{0.24} = 66.7\%$$

- Seeing the word "FREE" jumps the probability of the email being spam from **20% to 66.7%**.

Naïve Bayes Classification

Classification Problem



NAME	RANK	YEARS	TENURED
Tom	Assistant Prof	2	no
Merlisa	Associate Prof	7	no
George	Professor	5	yes
Joseph	Assistant Prof	7	yes

Bayes' Theorem: Basics

- Bayes' Theorem:

$$P(H \mid \mathbf{X}) = \frac{P(\mathbf{X} \mid H)P(H)}{P(\mathbf{X})} = P(\mathbf{X} \mid H) \times P(H) / P(\mathbf{X})$$

  - Let **X** be a data sample ("*evidence*"): class label is unknown
  - Let H be a *hypothesis* that X belongs to class C
  - Classification is to determine P(H|**X**), (i.e., *posteriori probability):* the probability that the hypothesis holds given the observed data sample **X**
  - P(H) (*prior probability*): the initial probability
  - P(**X**): probability that sample data is observed
  - P(**X**|H) (likelihood): the probability of observing the sample **X**, given that the hypothesis holds

Prediction Based on Bayes' Theorem

- Given training data **X**, *posteriori probability of a hypothesis* H, P(H|**X**), follows the Bayes' theorem

$$P(H \mid \mathbf{X}) = \frac{P(\mathbf{X} \mid H)P(H)}{P(\mathbf{X})} = P(\mathbf{X} \mid H) \times P(H) / P(\mathbf{X})$$

- Informally, this can be viewed as

  posteriori = likelihood x prior/evidence

- Predicts **X** belongs to C_i iff the probability $P(C_i|\mathbf{X})$ is the highest among all the $P(C_k|X)$ for all the *k* classes
- Practical difficulty: It requires initial knowledge of many probabilities, involving significant computational cost

Classification Is to Derive the Maximum Posteriori

- Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n-D attribute vector **X** = (x_1, x_2, …, x_n)
- Suppose there are *m* classes C_1, C_2, …, C_m.
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|\mathbf{X})$
- This can be derived from Bayes' theorem

$$P(C_i|\mathbf{X}) = \frac{P(\mathbf{X}|C_i)P(C_i)}{P(\mathbf{X})}$$

- Since P(X) is constant for all classes, only

$$P(C_i|\mathbf{X}) = P(\mathbf{X}|C_i)P(C_i)$$

  needs to be maximized

Naïve Bayes Classifier

- A simplified assumption: attributes are conditionally independent (i.e., no dependence relation between attributes):

$$P(\mathbf{X}|C_i) = \prod_{k=1}^{n} P(x_k|C_i) = P(x_1|C_i) \times P(x_2|C_i) \times ... \times P(x_n|C_i)$$

- This greatly reduces the computation cost: Only counts the class distribution

Naïve Bayes Classifier: Training Dataset

Class:
C1:buys_computer = ‘yes’
C2:buys_computer = ‘no’

Data to be classified:
X = (age <=30,
Income = medium,
Student = yes
Credit_rating = Fair)

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

Naïve Bayes Classifier: An Example

age	income	student	credit_rating	com
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

- $P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643

  P(buys_computer = "no") = 5/14= 0.357

- Compute $P(X|C_i)$ for each class

  P(age = "<=30" | buys_computer = "yes") = 2/9 = 0.222

  P(age = "<= 30" | buys_computer = "no") = 3/5 = 0.6

  P(income = "medium" | buys_computer = "yes") = 4/9 = 0.444

  P(income = "medium" | buys_computer = "no") = 2/5 = 0.4

  P(student = "yes" | buys_computer = "yes) = 6/9 = 0.667

  P(student = "yes" | buys_computer = "no") = 1/5 = 0.2

  P(credit_rating = "fair" | buys_computer = "yes") = 6/9 = 0.667

  P(credit_rating = "fair" | buys_computer = "no") = 2/5 = 0.4

- **X = (age <= 30 , income = medium, student = yes, credit_rating = fair)**

$P(X|C_i)$: P(X|buys_computer = "yes") = 0.222 x 0.444 x 0.667 x 0.667 = 0.044

P(X|buys_computer = "no") = 0.6 x 0.4 x 0.2 x 0.4 = 0.019

$P(X|C_i)*P(C_i)$: P(X|buys_computer = "yes") * P(buys_computer = "yes") = 0.028

P(X|buys_computer = "no") * P(buys_computer = "no") = 0.007

Therefore, X belongs to class ("buys_computer = yes")

Bayesian network (or belief network)

- A **Bayesian network** (or belief network) is a probabilistic graphical model that represents a set of random variables and their conditional dependencies using a Directed Acyclic Graph (DAG).
- It enables efficient reasoning under uncertainty by factorizing complex joint probability distributions into simpler local conditional distributions.

Core Structure & Components

- **Nodes**: Represent continuous or discrete random variables (e.g., events, measurements, or system states).
- **Directed Edges**: Arrows indicating direct conditional or causal dependencies. An arrow from node A to node B means A directly influences B.
- **Conditional Probability Tables (CPTs)**: Quantify the relationship between a node and its parents. Each node X_i maintains a local probability distribution given its immediate parents: $P\big(X_i \mid \text{Parents}(X_i)\big)$.

- The full joint probability distribution for a network of variables $X_1, X_2, \ldots, X_n$ is computed as the product of **all local conditional distributions**:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P\left(X_i \mid \text{Parents}(X_i)\right)$$

Key Features & Operations

- **Conditional Independence**: A node is conditionally independent of all its non-descendants given its parents. This property significantly reduces the number of parameters needed to model complex systems.
- **Probabilistic Inference**: Updates the probability distribution over unobserved target variables given observed evidence using Bayes' theorem:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

- **Parameter & Structure Learning**: The network structure and associated CPTs can either be designed manually by domain experts or learned automatically from data using algorithms like Maximum Likelihood Estimation (MLE) or Expectation-Maximization (EM).

Common Applications

- **Medical Diagnosis**: Evaluating the likelihood of underlying diseases given a combination of patient symptoms and clinical test outputs.
- **Risk Management & Finance**: Assessing credit default risk, detecting fraudulent transactions, and predicting market fluctuations.
- **Fault Diagnosis & Monitoring**: Identifying root causes of unexpected failures in industrial machinery and network systems.

Comparison: FJPD vs. Bayesian Network

- While the **Full Joint Probability Distribution (FJPD)** contains complete probabilistic information about a domain, storing and computing with it directly is computationally intractable for complex systems.
- A **Bayesian Network (BN)** is a structured, compact graph representation that factorizes the FJPD by exploiting local conditional independence properties.
- **Why We Need Bayesian Networks: Solving the Exponential Storage Problem**
- For $n$binary variables, a complete joint probability table requires storing $2^n - 1$independent entries:
  - **10 variables**: $2^{10} - 1 = 1{,}023$values (manageable).
  - **30 variables**: $2^{30} - 1 \approx 1.07 \times 10^9$values ($\sim 8.5$ GB of RAM for a single table).
  - **100 variables**: $2^{100} - 1 \approx 1.26 \times 10^{30}$values (exceeds total digital storage capacity on Earth).
- By enforcing local dependencies where each variable has at most $k$parents, a Bayesian Network reduces parameters to $n \cdot 2^k$. For $n = 100$and $k = 3$, a BN requires only $100 \cdot 2^3 = 800$values instead of 10^{30}.

- The full joint probability distribution for $n$random variables $X_1, X_2, \ldots, X_n$is expressed using the **Chain Rule of Probability**.

General Equation (Without Independence Assumptions)

$$P(X_1, X_2, \ldots, X_n) = P(X_1) \cdot P(X_2 \mid X_1) \cdot P(X_3 \mid X_1, X_2) \ldots P(X_n \mid X_1, X_2, \ldots, X_{n-1})$$

- In compact product notation:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P\left(X_i \mid X_1, \ldots, X_{i-1}\right)$$

- Every variable depends on all preceding variables in the ordering, requiring complete history at each step.

Factorized Equation (Bayesian Network / Markov Condition)

- When local conditional independence properties are applied via a Bayesian Network, the full joint probability distribution simplifies to:

$$P(X_1, X_2, \ldots, X_n) = \prod_{i=1}^{n} P\left(X_i \mid \text{Parents}(X_i)\right)$$

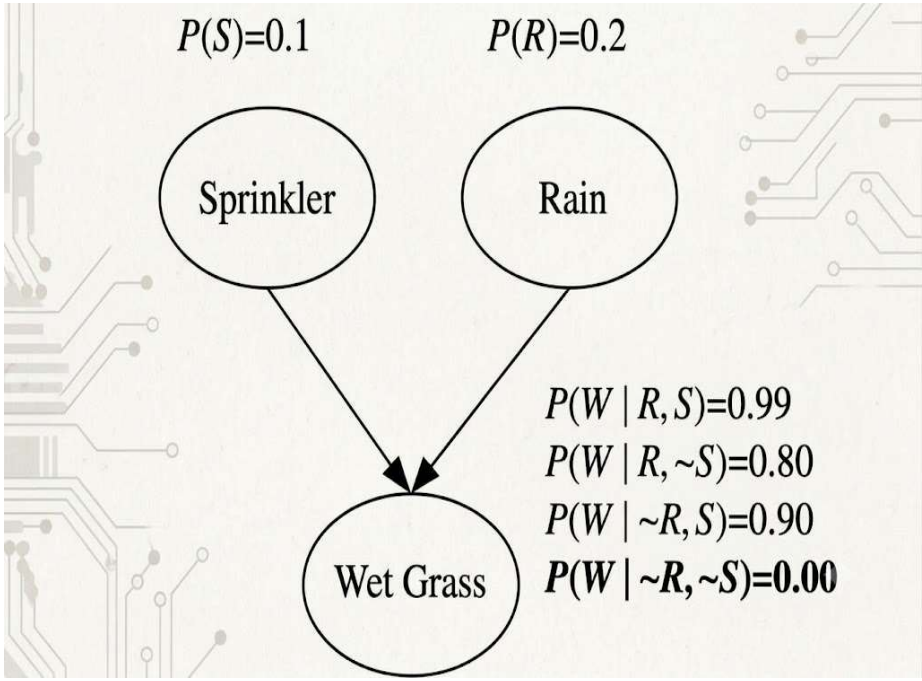
where $\text{Parents}(X_i) \subseteq \{X_1, \ldots, X_{i-1}\}$represents the set of direct parent nodes pointing to $X_i$in the network graph.

Example

A classic three-variable Bayesian network models the relationship between **Rain (R)**, **Sprinkler (S)**, and **Wet Grass (W)**, where whether the lawn is wet depends on whether it rained or if the sprinkler was turned on.

1. Network Structure (DAG)

- **Nodes**: $R \in \{T, F\}$, $S \in \{T, F\}$, $W \in \{T, F\}$
- **Edges**: Direct arrows $R \rightarrow W$ and $S \rightarrow W$
- **Factorization Formula**: $P(R, S, W) = P(R) \cdot P(S) \cdot P(W \mid R, S)$



Example Calculations

(a) Finding a Joint Probability

- To calculate the probability that **it rained**, **the sprinkler was off**, and **the grass is wet**:

$$P(R = \mathsf{T}, S = \mathsf{F}, W = \mathsf{T})$$
$$= P(R = \mathsf{T}) \cdot P(S = \mathsf{F}) \cdot P(W = \mathsf{T} \mid R = \mathsf{T}, S = \mathsf{F}) = 0.20 \times 0.90 \times 0.80$$
$$= 0.144 \quad (14.4\%)$$

(b) Probabilistic Inference / Exact Inference using Bayesian Network:

- If you observe that the grass is wet ($W = \mathrm{T}$), you can update your belief about whether it rained ($R = \mathrm{T}$) using Bayes' rule:

$$P(R = \mathrm{T} \mid W = \mathrm{T}) = \frac{P(R = \mathrm{T}, W = \mathrm{T})}{P(W = \mathrm{T})}$$

- **Calculate** $P(R = \mathrm{T}, W = \mathrm{T})$by marginalizing over S:
- $P(R = \mathrm{T}, W = \mathrm{T}) = P(\mathrm{T}, \mathrm{T}, \mathrm{T}) + P(\mathrm{T}, \mathrm{F}, \mathrm{T}) = (0.2 \times 0.1 \times 0.99) + (0.2 \times 0.9 \times 0.80) = 0.0198 + 0.144 = 0.1638$
- **Calculate total** $P(W = \mathrm{T})$by summing over all four joint configurations where $W = \mathrm{T}$:
- $P(W = \mathrm{T}) = P(\mathrm{R{=}T}, \mathrm{S{=}T}, \mathrm{W{=}T}) + P(R = \mathrm{T}, \mathrm{S{=}F}, \mathrm{W{=}T}) + P(R = \mathrm{F}, S = \mathrm{T}, W = \mathrm{T}) + P(R = \mathrm{F}, S = \mathrm{F}, W = \mathrm{T})$
- $P(W = \mathrm{T}) = 0.1638 + (0.8 \times 0.1 \times 0.90) + (0.8 \times 0.9 \times 0.00) = 0.1638 + 0.072 + 0 = 0.2358$
- **Final Posterior Probability**:

$$P(R = \mathrm{T} \mid W = \mathrm{T}) = \frac{0.1638}{0.2358} \approx 0.6946 \quad (69.46\%)$$

- Seeing wet grass increases the probability that it rained from an initial **20%** prior up to **69.46%**.

Important Observation:

Numerator: $P(R, W) = \sum_{s \in \{S, \sim S\}} P\,(R, s, W)$

- Expanding the sum for when the sprinkler is **ON (S)** and when the sprinkler is **OFF ($\sim S$)**:

$$P(R, W) = \underbrace{P(W \mid R, S) \cdot P(R) \cdot P(S)}_{\text{Scenario 1: Rained + Sprinkler ON}} + \underbrace{P(W \mid R, \sim S) \cdot P(R) \cdot P(\sim S)}_{\text{Scenario 2: Rained + Sprinkler OFF}}$$

Denominator:

- $P(W) = \sum_{r \in \{R, \sim R\}} \sum_{s \in \{S, \sim S\}} P\,(W \mid r, s) \cdot P(r) \cdot P(s)$

- Expanding all four combinations of R(Rain / No Rain) and S(Sprinkler ON / OFF):

$$P(W) = \underbrace{P(W \mid R, S) \cdot P(R) \cdot P(S)}_{\text{Scenario 1: Rained + Sprinkler ON}}$$

$$+ \underbrace{P(W \mid R, \sim S) \cdot P(R) \cdot P(\sim S)}_{\text{Scenario 2: Rained + Sprinkler OFF}}$$

$$+ \underbrace{P(W \mid \sim R, S) \cdot P(\sim R) \cdot P(S)}_{\text{Scenario 3: No Rain + Sprinkler ON}}$$

$$+ \underbrace{P(W \mid \sim R, \sim S) \cdot P(\sim R) \cdot P(\sim S)}_{\text{Scenario 4: No Rain + Sprinkler OFF}}$$

- $P(R \mid W) = \frac{P(R,W)}{P(W)} = \frac{\text{Scenario 1+Scenario 2}}{\text{Scenario 1+Scenario 2+Scenario 3+Scenario 4}}$

Why We Do It?

- **Conceptually: We want a result where S doesn't matter**
- In Numerator, Your target question is: *"What is the probability that it rained AND the grass is wet?"* Notice that this question makes no mention of the sprinkler. However, in the real world, the sprinkler still happened to be either ON or OFF. To find the total probability of Rain + Wet Grass regardless of the sprinkler, you must account for both possibilities and combine them. Here, only S is the hidden variable.
- Similarly, in denominator, To find the overall probability of seeing **Wet Grass (W)** on any given day, you must account for every possible combination of events that could cause or affect it. Here, both R and S are hidden variables.
- There are **4 mutually exclusive scenarios** in which the grass can end up wet:
  Both occurred: It rained AND the sprinkler was ON (R, S)
  Rain only: It rained AND the sprinkler was OFF $(R, \sim S)$
  Sprinkler only: It didn't rain AND the sprinkler was ON $(\sim R, S)$
  Neither occurred: It didn't rain AND the sprinkler was OFF $(\sim R, \sim S)$(even here, there's a 10%chance of dew or a spill)
- Because wet grass can happen under **any** of these 4 situations, you cannot ignore any combination. You must calculate the chance of wet grass in each scenario and add them all together to get the total $P(W)$.

Another Calculation

- To calculate $P(S \mid R, W)$, apply the definition of conditional probability: $P(S \mid R, W) = \frac{P(R,S,W)}{P(R,W)}$
- Numerator: $P(R, S, W) = P(W \mid R, S) \cdot P(R) \cdot P(S)$ **Calculate Numerator**: $0.95 \times 0.2 = \mathbf{0.19}$
- Denominator: $P(R, W) = P(W \mid R, S) \cdot P(R) \cdot P(S) \; + \; P(W \mid R, \sim S) \cdot P(R) \cdot P(\sim S)$ **Calculate Denominator**: $(0.95 \times 0.2) + (0.90 \times 0.8) = 0.19 + 0.72 = \mathbf{0.91}$
- **Combined Expanded Form:**
- $P(S \mid R, W) = \dfrac{\overbrace{P(W|R,S) \cdot P(R) \cdot P(S)}^{\text{Scenario 1: Rained + Sprinkler ON}}}{\underbrace{P(W|R,S) \cdot P(R) \cdot P(S)}_{\text{Scenario 1}} + \underbrace{P(W|R,\sim S) \cdot P(R) \cdot P(\sim S)}_{\text{Scenario 2: Rained + Sprinkler OFF}}}$
- $P(S \mid R, W) = \frac{0.19}{0.91} \approx \mathbf{0.2088} \quad (20.88\%)$

Application 2: Textbook Example

- **The Scenario :** You live in a quiet suburban home in Los Angeles. You have installed a burglar alarm that is reliable at detecting burglaries, but it can also be triggered by minor earthquakes. You have two neighbors, **John** and **Mary**, who have agreed to call you at work if they hear your alarm go off: **John** always calls when he hears the alarm, but he sometimes mistakes the telephone ringing for the alarm and calls anyway. **Mary** likes loud music and sometimes misses the alarm altogether.

The 5 Random Variables (All Binary: True/False**)**

- B: Burglary occurs.
- E: Earthquake occurs.
- A: Alarm sounds.
- J: John calls you at work.
- M: Mary calls you at work.

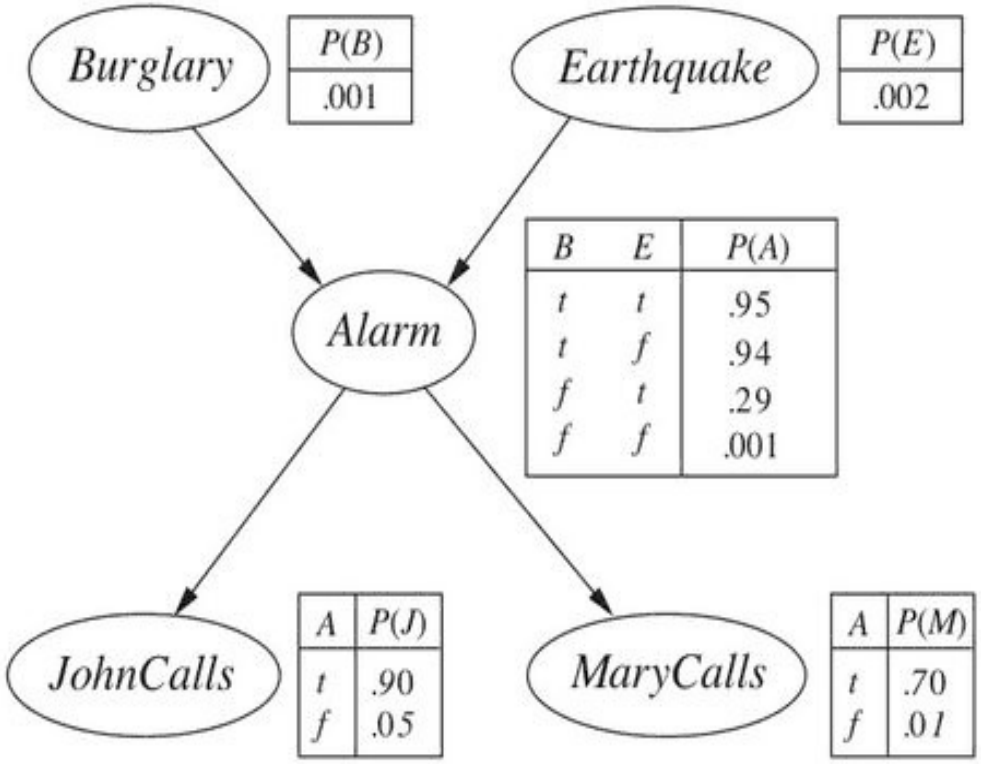


Figure 14.2 A typical Bayesian network, showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J, and M stand for *Burglary*, *Earthquake*, *Alarm*, *JohnCalls*, and *MaryCalls*, respectively.

- **Full Joint Factorization Formula:** Based on the direct edges in the DAG, the full joint probability distribution is factorized as:

$$P(B, E, A, J, M) = P(B) \cdot P(E) \cdot P(A \mid B, E) \cdot P(J \mid A) \cdot P(M \mid A)$$

Problem 1: Exact Joint Probability Query

- **Question**: *Calculate the probability that the alarm has sounded, but neither a burglary nor an earthquake has occurred, and both John and Mary call.*
- **Mathematical Form:**

$$P(J = \text{True}, M = \text{True}, A = \text{True}, B = \text{False}, E = \text{False})$$

$$\text{or} \quad P(j, m, a, \neg b, \neg e)$$

$$P(j, m, a, \neg b, \neg e) = P(\neg b) \cdot P(\neg e) \cdot P(a \mid \neg b, \neg e) \cdot P(j \mid a) \cdot P(m \mid a)$$

$$= 0.999 \times 0.998 \times 0.001 \times 0.90 \times 0.70$$

$$\approx 0.000628 \quad (\mathbf{0.0628}\%)$$

Problem 2: Diagnostic Inference Query

Question: *Calculate the posterior probability of a burglary given that both John and Mary call.*

Mathematical Form:

$$P(B = \text{True} \mid J = \text{True}, M = \text{True}) \quad \text{or} \quad P(b \mid j, m)$$

- By the definition of conditional probability:

$$P(b \mid j, m) = \frac{P(b, j, m)}{P(j, m)}$$

- In this network, the variables E(Earthquake) and A(Alarm) are not observed in the query $P(B \mid j, m)$. Therefore, they must be marginalized (summed out) across all possible combinations of their states: $E \in \{e, \neg e\}$and $A \in \{a, \neg a\}$.
- Using the factorized Bayesian Network formula:
  $$P(B, E, A, J, M) = P(B) \cdot P(E) \cdot P(A \mid B, E) \cdot P(J \mid A) \cdot P(M \mid A)$$
- The general marginalized formula for any state of B is:
  $$P(B, j, m) = \sum_{e \in \{e, \neg e\}} \sum_{a \in \{a, \neg a\}} P(B) \cdot P(e) \cdot P(a \mid B, e) \cdot P(j \mid a) \cdot P(m \mid a)$$

- **Fully Expanded Formula & Calculation for $P(b, j, m)$($B = \text{True}$)**

Expanding the double sum over all 4 combinations of (E, A):

$$P(b,j,m) = \underbrace{P(b) \cdot P(e) \cdot P(a \mid b,e) \cdot P(j \mid a) \cdot P(m \mid a)}_{\text{Term 1: } (e,a)}$$

$$+ \underbrace{P(b) \cdot P(e) \cdot P(\neg a \mid b,e) \cdot P(j \mid \neg a) \cdot P(m \mid \neg a)}_{\text{Term 2: } (e,\neg a)}$$

$$+ \underbrace{P(b) \cdot P(\neg e) \cdot P(a \mid b,\neg e) \cdot P(j \mid a) \cdot P(m \mid a)}_{\text{Term 3: } (\neg e,a)}$$

$$+ \underbrace{P(b) \cdot P(\neg e) \cdot P(\neg a \mid b,\neg e) \cdot P(j \mid \neg a) \cdot P(m \mid \neg a)}_{\text{Term 4: } (\neg e,\neg a)}$$

- **Substituting CPT Numerical Values:**
- $P(b) = 0.001$
- $P(e) = 0.002, \quad P(\neg e) = 0.998$
- $P(a \mid b, e) = 0.95, \quad P(\neg a \mid b, e) = 0.05$
- $P(a \mid b, \neg e) = 0.94, \quad P(\neg a \mid b, \neg e) = 0.06$
- $P(j \mid a) = 0.90, \quad P(j \mid \neg a) = 0.05$
- $P(m \mid a) = 0.70, \quad P(m \mid \neg a) = 0.01$

Term-by-Term Computation:

- **Term 1** (e, a):
$$0.001 \times 0.002 \times 0.95 \times 0.90 \times 0.70 = 0.0000011970$$
- **Term 2** $(e, \neg a)$:
$$0.001 \times 0.002 \times 0.05 \times 0.05 \times 0.01 = 0.0000000001$$
- **Term 3** $(\neg e, a)$:
$$0.001 \times 0.998 \times 0.94 \times 0.90 \times 0.70 = 0.0005910156$$
- **Term 4** $(\neg e, \neg a)$:
$$0.001 \times 0.998 \times 0.06 \times 0.05 \times 0.01 = 0.0000000299$$
- **Summing the 4 Terms:**
$P(b, j, m) = 0.0000011970 + 0.0000000001 + 0.0005910156 + 0.0000000299 = \mathbf{0.0005922426}$

- The complete expanded denominator $P(j, m)$is obtained by summing over all 8 joint states of the three unobserved variables: $B \in \{b, \neg b\}$, $E \in \{e, \neg e\}$, and $A \in \{a, \neg a\}$.

$$P(j, m) = \sum_{b' \in \{b, \neg b\}} \sum_{e' \in \{e, \neg e\}} \sum_{a' \in \{a, \neg a\}} P(b') \cdot P(e') \cdot P(a' \mid b', e') \cdot P(j \mid a') \cdot P(m \mid a')$$

- **Full 8-Term Expansion Formula**
- $P(j,m) = \text{Term 1} + \text{Term 2} + \text{Term 3} + \text{Term 4} + \text{Term 5} + \text{Term 6} + \text{Term 7} + \text{Term 8}$

$$P(j,m)\,\&= \underbrace{P(b)\cdot P(e)\cdot P(a \mid b,e)\cdot P(j \mid a)\cdot P(m \mid a)}_{\text{Term 1: } (b,e,a)}$$

$$+\ \underbrace{P(b)\cdot P(e)\cdot P(\neg a \mid b,e)\cdot P(j \mid \neg a)\cdot P(m \mid \neg a)}_{\text{Term 2: } (b,e,\neg a)}$$

$$+\ \underbrace{P(b)\cdot P(\neg e)\cdot P(a \mid b,\neg e)\cdot P(j \mid a)\cdot P(m \mid a)}_{\text{Term 3: } (b,\neg e,a)}$$

$$+\ \underbrace{P(b)\cdot P(\neg e)\cdot P(\neg a \mid b,\neg e)\cdot P(j \mid \neg a)\cdot P(m \mid \neg a)}_{\text{Term 4: } (b,\neg e,\neg a)}$$

$$+\ \underbrace{P(\neg b)\cdot P(e)\cdot P(a \mid \neg b,e)\cdot P(j \mid a)\cdot P(m \mid a)}_{\text{Term 5: } (\neg b,e,a)}$$

$$+\ \underbrace{P(\neg b)\cdot P(e)\cdot P(\neg a \mid \neg b,e)\cdot P(j \mid \neg a)\cdot P(m \mid \neg a)}_{\text{Term 6: } (\neg b,e,\neg a)}$$

$$+\ \underbrace{P(\neg b)\cdot P(\neg e)\cdot P(a \mid \neg b,\neg e)\cdot P(j \mid a)\cdot P(m \mid a)}_{\text{Term 7: } (\neg b,\neg e,a)}$$

$$+\ \underbrace{P(\neg b)\cdot P(\neg e)\cdot P(\neg a \mid \neg b,\neg e)\cdot P(j \mid \neg a)\cdot P(m \mid \neg a)}_{\text{Term 8: } (\neg b,\neg e,\neg a)}$$

Step-by-Step Substitution & Values for Each Term

- **Group 1: Burglary = True ($B = b$)**
- **Term 1** (b, e, a):

$$0.001 \times 0.002 \times 0.95 \times 0.90 \times 0.70 = \mathbf{0.0000011970}$$

- **Term 2** $(b, e, \neg a)$:

$$0.001 \times 0.002 \times 0.05 \times 0.05 \times 0.01 = \mathbf{0.0000000001}$$

- **Term 3** $(b, \neg e, a)$:

$$0.001 \times 0.998 \times 0.94 \times 0.90 \times 0.70 = \mathbf{0.0005910156}$$

- **Term 4** $(b, \neg e, \neg a)$:

$$0.001 \times 0.998 \times 0.06 \times 0.05 \times 0.01 = \mathbf{0.0000000299}$$

- **Group 2: Burglary = False ($B = \neg b$)**
- **Term 5** $(\neg b, e, a)$:

$$0.999 \times 0.002 \times 0.29 \times 0.90 \times 0.70 = \mathbf{0.0003650346}$$

- **Term 6** $(\neg b, e, \neg a)$:

$$0.999 \times 0.002 \times 0.71 \times 0.05 \times 0.01 = \mathbf{0.0000007093}$$

- **Term 7** $(\neg b, \neg e, a)$:

$$0.999 \times 0.998 \times 0.001 \times 0.90 \times 0.70 = \mathbf{0.0006281113}$$

- **Term 8** $(\neg b, \neg e, \neg a)$:

$$0.999 \times 0.998 \times 0.999 \times 0.05 \times 0.01 = \mathbf{0.0004980025}$$

- **Final Summation**

$$P(j, m) = 0.0000011970 + 0.0000000001 + 0.0005910156 + 0.0000000299$$

- $$+0.0003650346 + 0.0000007093 + 0.0006281113 + 0.0004980025$$
$$= \mathbf{0.0020841002}$$

- **Full Single-Fraction Equation Summary**
- Combining numerator and denominator into a single fully expanded Bayes' Rule equation:

$$P(b \mid j,m) = \frac{\sum_{e\in\{e,\neg e\}}\sum_{a\in\{a,\neg a\}} P\,(b)P(e)P(a \mid b,e)P(j \mid a)P(m \mid a)}{\sum_{b'\in\{b,\neg b\}}\sum_{e\in\{e,\neg e\}}\sum_{a\in\{a,\neg a\}} P\,(b')P(e)P(a \mid b',e)P(j \mid a)P(m \mid a)}$$

- Substituting the expanded values:

$$P(b \mid j,m) = \frac{0.0005922426}{0.0020841002} \approx \mathbf{0.2842} \quad (\mathbf{28.42}\%)$$

Problem 3

- **Question**: *What happens to the belief in a burglary if we receive additional news that a minor Earthquake (e) occurred at the same time?*
- **Mathematical Form:** $P(B = \text{True} \mid j, m, e)$
- **Step-by-Step Solution:** Since Earthquake is now observed as **True** (e), $E$is fixed, leaving only $A$to be marginalized.
- **For** $B = \text{True}$**(**e **is fixed to True)**:

$$P(b, e, j, m) = \sum_{a} P(b) \cdot P(e) \cdot P(a \mid b, e) \cdot P(j \mid a) \cdot P(m \mid a)$$

$$= \underbrace{0.001 \times 0.002 \times 0.95 \times 0.90 \times 0.70}_{\text{Alarm = True}} + \underbrace{0.001 \times 0.002 \times 0.05 \times 0.05 \times 0.01}_{\text{Alarm = False}}$$

$$= 0.0000011970 + 0.0000000001 = \mathbf{0.0000011971}$$

- **For** $B = \text{False}$**(**e **is fixed to True)**:

$$P(\neg b, e, j, m) = \sum_{a} P(\neg b) \cdot P(e) \cdot P(a \mid \neg b, e) \cdot P(j \mid a) \cdot P(m \mid a)$$

$$= \underbrace{0.999 \times 0.002 \times 0.29 \times 0.90 \times 0.70}_{\text{Alarm = True}} + \underbrace{0.999 \times 0.002 \times 0.71 \times 0.05 \times 0.01}_{\text{Alarm = False}}$$

$$= 0.0003650346 + 0.0000007093 = \mathbf{0.0003657439}$$

- $P(b \mid j, m, e) = \frac{0.0000011971}{0.0000011971+0.0003657439}$

$$= \frac{0.0000011971}{0.0003669410} \approx \mathbf{0.00326} \quad (\mathbf{0.33}\%)$$